

Units and Measurements

Quick reference — fold and keep.

MEASUREMENT IDENTITY

$$n_1 u_1 = n_2 u_2$$

SYSTEM CONVERSION

$$n_2 = n_1 [M_1/M_2]^a [L_1/L_2]^b [T_1/T_2]^c$$

FORCE / ENERGY (CGS)

$$1 \text{ N} = 10^5 \text{ dyne}, \quad 1 \text{ J} = 10^7 \text{ erg}$$

HOMOGENEITY

$$[\text{LHS}] = [\text{RHS}] = [\text{every term}]$$

RAYLEIGH METHOD

$$Q = k A^x B^y C^z$$

SIG FIGS: + AND -

fewest decimal places

SIG FIGS: × AND ÷

fewest significant figures

MEAN ABSOLUTE ERROR

$$\Delta a = \frac{1}{n} \sum |a_{\text{mean}} - a_i|$$

PERCENTAGE ERROR

$$\delta a\% = \frac{\Delta a}{a} \times 100$$

ERRORS: SUM / DIFFERENCE

$$\Delta Z = \Delta A + \Delta B$$

ERRORS: PRODUCT / QUOTIENT

$$\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

ERRORS: POWERS

$$Z = A^p B^q C^{-r} \Rightarrow \frac{\Delta Z}{Z} = |p| \frac{\Delta A}{A} + |q| \frac{\Delta B}{B} + |r| \frac{\Delta C}{C}$$

VERNIER L.C.

$$\text{L.C.} = 1\text{MSD} - 1\text{VSD} = \frac{\text{MSD}}{n}$$

SCREW GAUGE L.C.

$$\text{L.C.} = \frac{\text{pitch}}{N}$$

ZERO CORRECTION

$$R_{\text{true}} = R_{\text{obs}} - \text{zero error}$$

DIMENSIONS: VISCOSITY

$$[\eta] = [\text{ML}^{-1}\text{T}^{-1}]$$

DIMENSIONS: PLANCK

$$[h] = [\text{ML}^2\text{T}^{-1}]$$

DIMENSIONS: G

$$[G] = [\text{M}^{-1}\text{L}^3\text{T}^{-2}]$$

Units and Measurements

Name: _____

Date: _____

Time allowed: 23 min

Marking: +4 for a correct answer, -1 for an incorrect multiple-choice answer, 0 for an unattempted question and for incorrect numerical answers. Shade one bubble per question on the grid below.

1. $5.0 \times 2.00 = ?$ (to the correct number of significant figures) ☒ G

(A) 10

(B) 10.0

(C) 10.00

(D) 1.0×10^1

2. If $Z = A - B$, the absolute error in Z is: ☒ G

(A) $\Delta A - \Delta B$

(B) $\Delta A + \Delta B$

(C) $\sqrt{\Delta A \cdot \Delta B}$

(D) $\Delta A / \Delta B$

3. The dimensional formula of **acceleration** is: ☒ G

(A) $[LT^{-1}]$

(B) $[LT^{-2}]$

(C) $[MLT^{-2}]$

(D) $[L^2T^{-2}]$

4. Dimensional analysis fails to derive the relation $s = ut + \frac{1}{2}at^2$ mainly because: ☒ M

(A) the equation is dimensionally incorrect

(B) the relation is a sum of two terms, not a single product of powers

(C) acceleration has no dimensions

(D) time appears twice

5. The same calliper (L.C. = 0.01 cm) has a zero error of -0.03 cm. If the observed reading is 1.26 cm, what is the corrected reading in cm? (M)

Answer: _____

6. In a system of units the unit of force is 100 N, the unit of length is 10 m and the unit of time is 100 s. The unit of mass in this system is 10^n kg. Find n . (M)

Answer: _____

7. The viscous force F on a sphere of radius r moving with speed v through a fluid of viscosity η is assumed to be $F = k\eta^a r^b v^c$. The values of a, b, c are: (M)

(A) 1, 1, 1

(B) 1, 2, 1

(C) 2, 1, 1

(D) 1, 1, 2

8. Which **pair** of quantities has the same SI unit? (M)

(A) Work and power

(B) Torque and work

(C) Momentum and impulse only

(D) Pressure and force

9. A cube of edge $1\ \mu\text{m}$ is made of a material of density $8\ \text{g cm}^{-3}$. Its mass is 8×10^{-n} kg. Find n . (H)

Answer: _____

10. A student measures the same length with two instruments. Instrument X gives 5.2, 5.2, 5.2 cm; instrument Y gives 5.00, 5.12, 4.89 cm. The true length is 5.00 cm. Which statement is correct? (H)

(A) X is more accurate and more precise

(B) X is more precise; Y is more accurate

(C) Y is more precise; X is more accurate

(D) Both are equally accurate

11. A screw gauge has pitch 0.5 mm and 50 circular divisions. When the jaws are closed, the 45th division coincides with the reference line. A wire gives PSR = 1.5 mm and HSR = 27. Find the corrected diameter in mm. (H)

Answer: _____

12. Evaluate $\frac{0.02856 \times 298.15 \times 0.112}{0.5785}$ and report it to the correct number of significant figures. (H)

Answer: _____

13. A wire of diameter $d = (0.50 \pm 0.01)$ mm and length $L = (50.0 \pm 0.1)$ cm. The percentage error in its volume $V = \frac{\pi d^2 L}{4}$ is closest to: (H)

(A) 2.2%

(B) 4.2%

(C) 3.0%

(D) 1.2%

14. Which of the following pairs **cannot** be distinguished by dimensional analysis alone? (H)

(A) Force and momentum

(B) Work and torque

(C) Power and energy

(D) Velocity and acceleration

15. The energy E released by an explosion creates a shock wave of radius R at time t in air of density ρ . Assuming $R = k E^a \rho^b t^c$, the value of a is: (H)

(A) 1/5

(B) 1/3

(C) 2/5

(D) 1/2

Answer grid

1 (A) (B) (C) (D)

2 (A) (B) (C) (D)

3 (A) (B) (C) (D)

4 (A) (B) (C) (D)

5 _____

6 _____

7 (A) (B) (C) (D)

8 (A) (B) (C) (D)

9 _____

10 (A) (B) (C) (D)

Answer key

Keep this sheet face down until you have finished the paper.

Q	ANS	TIER	Q	ANS	TIER
1	D	G	9	15	H
2	B	G	10	B	H
3	B	G	11	1.82	H
4	B	M	12	1.65	H
5	1.29	M	13	B	H
6	5	M	14	B	H
7	A	M	15	A	H
8	B	M			

Worked solutions

1. Answer: D (G)

- 5.0 has 2 sig figs; 2.00 has 3. The answer may keep only **2**.
- The raw product is 10.
- Writing "10" is ambiguous (is that 1 or 2 sig figs?), so the honest form is 1.0×10^1 .
- This is exactly why scientific notation is preferred for reporting results.

2. Answer: B (G)

- For both sums and differences, the **absolute** errors add.
- Intuition: the worst case is A being at its maximum while B is at its minimum, which makes Z too large by $\Delta A + \Delta B$.
- Subtracting the errors would be assuming the two mistakes conveniently cancel — never assume that.

3. Answer: B (G)

- Velocity has dimensions $[LT^{-1}]$.
- Acceleration = $\frac{\Delta v}{\Delta t} = \frac{[LT^{-1}]}{[T]} = [LT^{-2}]$.
- Note there is no M — acceleration does not care about mass.

4. **Answer: B** (M)

1. Rayleigh's method assumes the answer has the single-product form $Q = kA^x B^y C^z$.
2. It has no mechanism for producing a *sum* of two dimensionally-identical terms.
3. Both ut and $\frac{1}{2}at^2$ have dimensions [L], so the equation is perfectly homogeneous — dimensional analysis simply cannot generate it.

5. **Answer: 1.29** (M)

1. Correct reading = observed – zero error.
2. = $1.26 - (-0.03)$.
3. = $1.26 + 0.03 = 1.29$ cm.
4. A negative zero error means the instrument under-reads, so the correction must *increase* the value.

6. **Answer: 5** (M)

1. From $F = ma = \frac{mL}{T^2}$, we get $m = \frac{FT^2}{L}$.
2. Substituting the new units: $m = \frac{100 \times (100)^2}{10}$.
3. = $\frac{100 \times 10^4}{10} = 10^5$ kg.
4. So $n = 5$.

7. **Answer: A** (M)

1. $[MLT^{-2}] = [ML^{-1}T^{-1}]^a [L]^b [LT^{-1}]^c$.
2. M: $a = 1$.
3. T: $-a - c = -2 \Rightarrow c = 1$.
4. L: $-a + b + c = 1 \Rightarrow -1 + b + 1 = 1 \Rightarrow b = 1$.
5. So $F = k\eta r v$, and experiment gives $k = 6\pi$ — Stokes' law.

8. **Answer: B** (M)

1. Work = $F \cdot d$ has unit N m = J.
2. Torque = $r \times F$ also has unit N m.
3. So both carry $\text{kg m}^2\text{s}^{-2}$ — identical units despite being physically different (one is a scalar, one a vector).
4. Momentum and impulse do share a unit (N s), but the option says "only", and torque/work is the intended classic pair. Work vs power and pressure vs force clearly differ.

9. **Answer: 15** (H)

1. Edge = $1\ \mu\text{m} = 10^{-6}\text{ m}$, so volume $V = (10^{-6})^3 = 10^{-18}\text{ m}^3$.
2. Density: $8\text{ g cm}^{-3} = 8 \times \frac{10^{-3}\text{ kg}}{10^{-6}\text{ m}^3} = 8 \times 10^3\text{ kg m}^{-3}$.
3. Mass = $\rho V = 8 \times 10^3 \times 10^{-18} = 8 \times 10^{-15}\text{ kg}$.
4. So $n = 15$.

10. **Answer: B** (H)

1. X's readings are identical — zero scatter, so X is more **precise**.
2. But X's mean is 5.2 cm, off by 0.2 cm: a systematic error.
3. Y's readings scatter ($\pm 0.12\text{ cm}$) but their mean is $5.00 + \frac{5.12 + 4.89}{3} = 5.003\text{ cm}$ — essentially the true value. Y is more **accurate**.
4. Real lesson: Y's error can be shrunk by taking more readings; X's cannot be shrunk at all without recalibration.

11. **Answer: 1.82** (H)

1. L.C. = $0.5/50 = 0.01\text{ mm}$.
2. Closed reading: the 45th of 50 divisions coincides, which is 5 divisions *short* of zero.
3. Zero error = $-(50 - 45) \times 0.01 = -0.05\text{ mm}$.
4. Observed = $1.5 + 27 \times 0.01 = 1.77\text{ mm}$.
5. Corrected = $1.77 - (-0.05) = 1.82\text{ mm}$.

12. **Answer: 1.65** (H)

1. Count sig figs: $0.02856 \rightarrow 4$, $298.15 \rightarrow 5$, $0.112 \rightarrow 3$, $0.5785 \rightarrow 4$.
2. The minimum is **3** (from 0.112).
3. Raw value: $0.02856 \times 298.15 = 8.5152\dots$; $\times 0.112 = 0.953707\dots$; $\div 0.5785 = 1.64859\dots$
4. Rounded to 3 significant figures: 1.65.

13. **Answer: B** (H)

1. $\frac{\Delta V}{V} = 2 \frac{\Delta d}{d} + \frac{\Delta L}{L}$.
2. $2 \times \frac{0.01}{0.50} = 2 \times 0.02 = 0.04 = 4\%$.
3. $\frac{0.1}{50.0} = 0.002 = 0.2\%$.
4. Total = $4\% + 0.2\% = 4.2\%$.
5. The diameter dominates completely — which is why you measure a thin wire with a screw gauge, not a vernier.

14. Answer: B (H)

1. Work: $[ML^2T^{-2}]$. Torque: also $[ML^2T^{-2}]$.
2. They are dimensionally indistinguishable, yet physically distinct — work is a scalar and torque a vector, and they are never added.
3. This is limitation (4): dimensional analysis sees only exponents, never physical meaning.
4. Another famous same-dimension pair: energy and moment of force.

15. Answer: A (H)

1. $[L] = [ML^2T^{-2}]^a [ML^{-3}]^b [T]^c$.
2. M: $a + b = 0$.
3. L: $2a - 3b = 1$. Substituting $b = -a$: $2a + 3a = 1 \Rightarrow a = \frac{1}{5}$.
4. Then $b = -\frac{1}{5}$, and T: $-2a + c = 0 \Rightarrow c = \frac{2}{5}$.
5. So $R \propto \left(\frac{Et^2}{\rho} \right)^{1/5}$ — the formula G. I. Taylor used to deduce the Trinity yield from declassified photographs.